

17. Disorders of Calcium, Magnesium, and Phosphate Balance · assets 1-9 / 13

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Box_e17_1

Box 17.2 Clinical Features of Hypercalcaemia

General
Malaise, tiredness, weakness
Neuropsychiatric
Impaired concentration, loss of memory, headache, drowsiness, lethargy, incontinence, confusion, irritability, depressed mood, paranoia, hallucinations, ataxia, speech difficulties, visual disturbances, deafness (calcification of eardrum), pruritus, mental relaxation (relaxation), stupor, coma
Neuromuscular
Muscle weakness, hyporeflexia or absent reflexes, hypotonia, myalgia, entrapment, bone pain, joint effusion, chondrocalcinosis, rhabdomyolysis
Gastrointestinal
Loss of appetite, dry mouth, thirst, polydipsia, nausea, vomiting, constipation, abdominal pain, weight loss, acute pancreatitis (calcification), peptic ulcer, acute gastric outlet
Kidney
Proteinuria, nocturia, nephrocalcinosis, nephrolithiasis, interstitial nephritis, acute kidney injury, and chronic kidney disease
Cardiovascular
Arrhythmias, bradycardia, first-degree heart block, short Q-T interval, bundle branch block, arrest (rare), hypertension, vasculopathy
Metastatic Calcification
Band keratopathy, red eye syndrome, conjunctival calcification (neoplastic), vascular calcification, pruritus

Box_e17_2

eBox 17.3 Clinical Features of Hypocalcemia

Neuromuscular Irritability

- General fatigability and muscle weakness
- Paresthesias, numbness
- Circumoral and peripheral extremity tingling
- Muscle twitching and cramping
- Tetany, carpal/pedal spasms
- Chvostek sign, Trousseau sign
- Laryngeal and bronchial spasms

Altered Central Nervous System Function

- Emotional disturbances – irritability, depression
- Altered mental status, coma
- Tonic-clonic seizures
- Papilledema, pseudotumor cerebri
- Cerebral calcifications

Cardiovascular

- Lengthening of the QTc interval
- Dysrhythmias
- Hypotension
- Congestive heart failure

Dermatologic and Ocular

- Dry skin, coarse hair, brittle nails
- Cataracts

Box_e17_3

Box 17.4 Causes of Hypocalcemia	
Inherited and Genetic Syndromes With Hypoparathyroidism • Rare mutations isolated congenital hypoparathyroidism • Autosomal recessive hypoparathyroidism with activating mutation of the <i>CaSR</i> (OMIM 145000) • Idiopathic syndrome (OMIM 185000) • Other forms of familial hypoparathyroidism	Vitamin D Resistance • Pseudovitamin D deficiency rickets (vitamin D-dependent rickets type 1) • Vitamin D-resistant rickets (vitamin D-dependent rickets type 2)
Acquired Hypoparathyroidism With Resistance to PTH Action • Pseudohypoparathyroidism types 1a, 1b, and 2 • Hypoparathyroidism syndromes	Miscellaneous Causes • Hypophosphatemia • Phosphate repletion caused by acute or chronic kidney disease • Excess phosphate absorption caused by enemas, oral supplements • Massive phosphate absorption caused by tumor lysis or crush injury
Acquired Hypoparathyroidism, Inadequate PTH Production • Decreased destruction of the parathyroid glands • Postoperative • Autoimmune-mediated or with multiple endocrine dysfunction	Drugs • Foscarnet • Bisphosphonate therapy (especially in patients with vitamin D deficiency) • Denosumab
• Acquired antibodies against <i>CaSR</i> • Pseudohypoparathyroidism type 1 (OMIM 243200 and 607300)	Rapid Transfusion of Large Volumes of Citrate-Containing Blood
• Metabolic and infiltrative diseases • Deposition of heavy metals—iron overload, copper overload • Reversible inhibition of PTH secretion • Severe hypomagnesemia • Hypomagnesemia	Acute Critical Illness (Multiple Contributing Causes)
Inadequate Vitamin D Nutrition • Vitamin D deficiency—racial, lack of sunlight exposure • Malabsorption • End-stage liver disease and cirrhosis • Chronic kidney disease	Hungry Bone Syndrome, Reallocating Calcium • Postthyroidectomy for Graves disease • Postoperative hypoparathyroidism
	Osteoblastic Metastases
	Acute Pancreatitis
	Rhabdomyolysis
	Substances Interfering With Laboratory Assay for Total Calcium • Glibenclamide can in contrast agents again during MRI, MRA

CaGR, calcium-sensing receptor; AMP, magnetic resonance imaging; AMP, magnetic resonance angiography; OMIM, Online Mendelian Inheritance in Man; PTH, parathyroid hormone.

Box_e17_4

Box 17.5 Causes of Hyperphosphatemia

Decreased Kidney Excretion of Phosphorus

Chronic kidney disease stages 3-5
Acute kidney injury
Hypoparathyroidism, pseudohypoparathyroidism
Acromegaly
Tumoral calcinosis
• Growth factor (FGF-23) inactivating gene mutation
• GALNT3 mutation with aberrant FGF-23 glycosylation
• KLOTHO inactivating mutation with FGF-23 resistance
• Bisphosphonates

Exogenous Phosphorus Administration

Ingestion of phosphate, phosphate-containing enemas
Excessive phosphate delivery

Redistribution of Phosphorus (Intracellular to Extracellular Shift)

Respiratory acidosis, metabolic acidosis
Tumor lysis syndrome
Rhabdomyolysis
Hemolytic anemia
Catabolic state

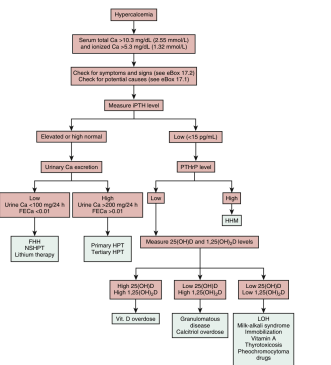
Pseudohyperphosphatemia

Hyperbilirubinemia
Hyperlipidemia
Hypertriglyceridemia
Medications
• Liposomal amphotericin B
• Rice tissue plasma activator
• Heparin
Hemolysis of blood specimen

Box_e17_5

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Box_e17_6



Fig_17_1

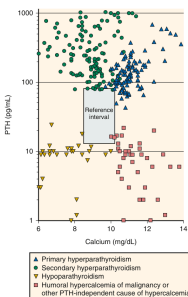


Fig. 17.2 Relationship between total serum calcium and intact parathyroid hormone (PTH) concentrations. Values for patients with primary hyperparathyroidism, secondary hyperparathyroidism, humoral hypercalcemia of malignancy, or other PTH-independent causes of hypercalcemia, and hypoparathyroidism are plotted. The rectangle represents the normal reference range for the assays. [From O'Neill S, Gordon C, Guo R, et al. Multivariate analysis of clinical, demographic, and laboratory data for classification of patients with disorders of calcium homeostasis. *Am J Clin Pathol*. 2011;135:100-107.]

Fig_17_2

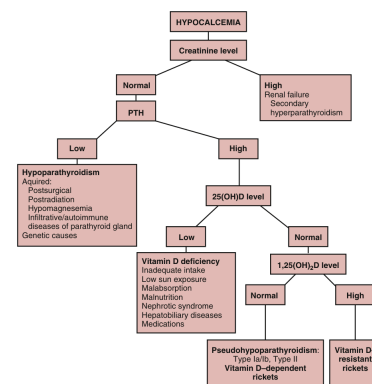


Fig. 17.3 Algorithm for the evaluation of hypocalcemia. PTH, Parathyroid hormone.

Fig_17_3

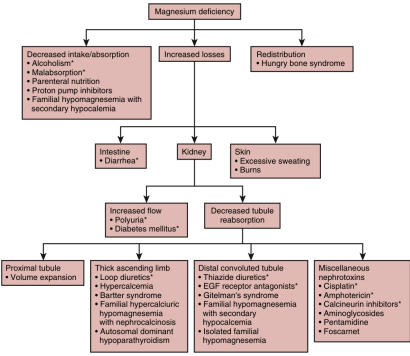


Fig. 17.4 Causes of magnesium deficiency. EGF, Epidermal growth factor.

Fig_17_4

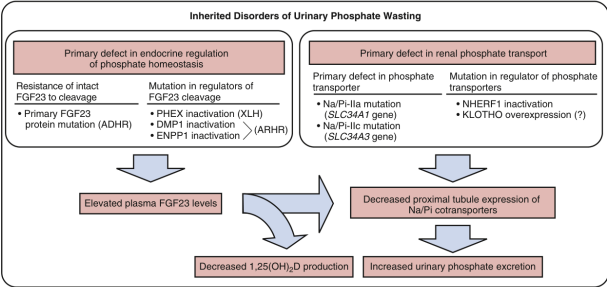


Fig. 17.5 Summary of the inherited disorders of urinary phosphate wasting. Inherited diseases characterized by phosphaturia and hypophosphatemia can occur from a defect in endocrine pathways involved in the systemic regulation of phosphate homeostasis or from a direct mutation in local regulators of renal phosphate transport. ADHR, Autosomal dominant hypophosphatemic rickets; ARHR, autosomal recessive hypophosphatemic rickets; DMP1, dentin matrix protein 1; ENPP1, ectonucleotide pyrophosphatase/phosphodiesterase 1; FGF23, fibroblast growth factor 23; Na/Pi-IIa, type IIa sodium-phosphate cotransporter; Na/Pi-IIc, type IIc sodium-phosphate cotransporter; NHERF1, sodium-hydrogen exchanger regulatory factor; PHEX, phosphate-regulating gene with homology to endopeptidases on the X chromosome; XLH, X-linked hypophosphatemic rickets.

Fig_17_5

Table 17.1 Pharmacologic Therapy for Hypercalcemia ^a		
Intervention or Calcuretic	Dose	Adverse Effects
Intermittent saline	200–500 mL/h, depending on patient's cardiovascular and kidney status	Competitive heart failure
Furosemide	20–40 mg IV (after hydration has been achieved)	Dehydration, hypokalemia, hyponatremia
First-Line Medications		
IV bisphosphonates ^b		
Pamidronate	60–90 mg IV over 2 hours in 50–200 mL saline solution or 9% dextrose in water	Acute kidney injury, transient flu-like syndrome with aches, chills, and fever
Zoledronate	4 mg IV over 15 minutes in 50 mL of saline solution or 5% dextrose in water	Acute kidney injury, transient flu-like syndrome with aches, chills, and fever
Second-Line Medications		
Glucocorticoids ^c	Example: prednisone, 60 mg orally daily, for 10 days	Potential interference with chemotherapy; hypokalemia, hyperglycemia, hypertension, Cushing syndrome, immunosuppression
Mithramycin	Single dose of 25 µg/kg of body weight over 4–6 hours in saline	Thrombocytopenia, platelet aggregation defect, anemia, leukopenia, hepatitis, kidney failure ^d
Calcitonin	4–8 U/kg subcutaneously or intramuscularly every 12 hours	Flushing, nausea, escape phenomenon
Gallium nitrate	100–200 mg/m ² of body surface area IV given continuously over 24 hours for 5 days	Acute kidney injury
Denosumab ^e	120 mg on days 1, 8, 15, and 29 and every 4 weeks	Hypocalcemia, hypophosphatemia, osteonecrosis of the jaw, atypical femoral fracture

^aMany recommendations in this table are based on historical precedent and common practice rather than on randomized clinical trials. There are data from randomized trials comparing bisphosphonates to the other agents listed and to one another.

^bPamidronate and zoledronate are approved by the U.S. Food and Drug Administration (FDA); zoledronate and clodronate are available in continental Europe, the United Kingdom, and elsewhere. Bisphosphonates should be used with caution, if at all, when the serum creatinine level exceeds 2.5 to 3.0 mg/dL (221.0–265.2 µmol/L).

^cHydrocortisone is generally used at a dose of 60 mg, but the 60-mg dose may be used to treat patients of small stature or those with kidney impairment or mild hypercalcemia.

^dThese drugs have a slow onset of action; as compared with bisphosphonates, approximately 4 to 10 days are required for a response.

^eThese effects have been reported in association with higher-dose regimens used to treat testicular cancer (50 µg/kg body weight/day over a period of 5 days) and in patients receiving multiple doses of 25 µg/kg; they are not expected to occur with a single dose of 25 µg/kg unless preexisting liver, kidney, or hematologic disease is present.

Approved by the FDA for the prevention of skeletal-related events in patients with bone metastases from solid tumors; used in an open-label fashion as a rescue therapy if bisphosphonates are not effective.

Table_17_1

Table 17.2 Genetic Syndromes With Hypoparathyroidism					
Reference	Syndrome	Other Clinical Features	Process Affected	Inheritance	Gene Mutated
Ding et al. ¹⁰¹	Familial isolated hypoparathyroidism type 2		Parathyroid gland development	AR	CCND2
Bowl et al. ¹⁰²				X-linked	SOX3
Parfitt and Thakker ¹⁰³	Familial isolated hypoparathyroidism type 1		PTH gene mutation affecting its synthesis	AR	Progn-PTH splice site
Arnold et al. ¹⁰⁴	Autosomal dominant hypocalcemia with hypercalciuria type 1		Calcium sensing	AD	Progn-PTH signal peptide
Pollak et al. ¹⁰⁵				AD	CaSR
Neale et al. ¹⁰⁶	Autosomal dominant hypocalcemia with hypercalciuria type 2			AD	GNA11
Yagi et al. ¹⁰⁷				Sporadic or AD	22q11 deletion (including TBX1)
Vissers et al. ¹⁰⁸	Charge	Cardiac anomalies, abnormal facial features, aplasia, cleft palate	Defective third and fourth branchial pouch development	Sporadic or AD	CHD7
Kobayashi et al. ¹⁰⁹				Sporadic or AD	SEMA3F
Van Esch et al. ¹¹⁰	HDR	Cystic degeneration, skeletal hypoplasia, and acromioclavicular dysplasia	Parathyroid development	AD	GATA3 transcription factor
Parfitt et al. ¹¹¹	Kenny-Caffey, Sengco-Sakai	Microcephaly, mental retardation, growth failure, osteoporosis		AR	TAC3 (signature for tubulin folding)
Heuland et al. ¹¹²	Autosomal recessive hypocalcemia with hypercalciuria (ARHCD)			AR	ARE (autosome translocator regulator)

AD, Autosomal dominant; AR, autosomal recessive; CaSR, calcium-sensing receptor; CHD7, Chromosome 17 Inheritance in Man; HDR, HDR.

Table_17_2